

Exploring the Novel Real-Time Pipeline of Nickelodeon's Max & the Midknights

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Figure 1: Exploring the Novel Real-Time Pipeline of Nickelodeon's Max & the Midknights

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1 Overview

Come hear about the creation of Nickelodeon's hit new animated action/adventure series Max & the Midknights! The show's Supervising Producer and CG Supervisor, along with production partner Xentrix's Head of Pipeline and Creative Director, will share a behind-the-scenes look at the challenges they faced creating this ambitious cinematic CG show with its hand-made and stop-motion look and feel.

You will learn how Max's design-forward pipeline, built upon a custom set of tools developed in Unreal, enabled a new agile story process that blends the best qualities of live-action and animated filmmaking. Unlike the "traditional" storyboard-based TV approach, Max's visualization artists shoot on final models in 3D

then deliver dailies to editors who leverage their live-action expertise to assemble a full episode for review. Once noted and updated, the visualization pass goes to the storyboard artists, who—thanks to the true 3D nature of the visualization work—can focus exclusively on character performance with the confidence that all of their compositions will be fully-reproducible in 3D. This hybridized process obviates the need for a downstream blocking/layout pass because the cameras and animations used to make the animatic are exported from Unreal and delivered along with the next-level animatic to Max's Bangalore-based production partner Xentrix Studios.

In addition to Max's unique story pipeline, you will also hear about the tools and techniques developed collaboratively between Nickelodeon and Xentrix to permit character animation in Maya with final rendering/delivery in Unreal, including a seamless alembic- and FBX-based animation transfer approach and automatic shot creation in Unreal from Maya data. The team will also highlight the creative power of real-time iteration for improving cameras, shading, and lighting later in the process than permitted in the typical pipeline, and how they were able to get broadcast-ready final pixels straight out of Unreal.